Supplementary Material for IdentiFace

I. Technical Details on Modality Fusion

In IdentiFace, we proposed a multi-modal conditional input design, which includes two conditional images. To align with the uni-modal image input of a uni-branch ControlNet [16], we fused the images together before injecting into it.

Observing that LQ images primarily retain low-frequency structural information while sketches capture high-frequency edge details, we designed a lightweight Modality Fusion Module to combine dual-modal inputs in the frequency domain. Through this module, we transformed dual-modal images into a uni-modal input. The visualization of this module and frequency analysis is shown in Fig. 1.

The module contains a frequency mixer and a residual correction CNN. Given inputs $I_{LQ}, I_{Sk} \in \mathbb{R}^{B\times C\times H\times W}$, which respectively refer to LQ and sketch image, the frequency mixer transforms them into the frequency domain via 2D FFT, and uses trainable low-pass filter masks $\mathbf{M}_{\phi}^{\text{low}}$ to partition the frequency spectrum. The fused representation I_{Fused} is obtained by inverse FFT:

$$I_{Fuse} = \Re\Big(\mathcal{F}^{-1}(\text{Lerp}(\widehat{I_{Sk}}, \widehat{I_{LQ}}, \mathbf{M}_{\phi}^{\text{low}})\Big)\,, \tag{1}$$

where

$$\widehat{I_{LQ}} = \mathcal{F}(I_{LQ}), \qquad \widehat{I_{Sk}} = \mathcal{F}(I_{Sk}). \tag{2}$$

To refine spatial details and correct minor misalignments, the residual correction CNN $g_{\phi'}$ processes the concatenated raw inputs to predict a residual map

$$I_{Res} = g_{\phi'}([I_{LQ}, I_{Sk}])\,. \tag{3}$$

The final conditional input is:

$$I_{Cond} = I_{Fuse} + \alpha \cdot I_{Res}. \tag{4}$$

with $\alpha = 0.2$ balancing two branches.

For our IdentiFace instances ID-SD and ID-PA, the module fuses images respectively in RGB Channels and latent-space channels, as the official ControlNet implementation [2] of PixArt-α receives latent-space conditional images.

The module is optimized jointly with ControlNet but remains frozen during initial steps to stabilize training. Convolutional weights ϕ' are zero-initialized, while frequency mixer parameters ϕ are initialized from prior knowledge.

Additionally, it is worth noticing that our proposed fusion module is just a **feasible** method for combining conditional images to directly process them by ControlNet. other approaches including bandpass filtering, pixel-level layer stacking (I_{Sk} over I_{LQ}) are also feasible for a similar I_{Cond} result.

II. Dataset Details

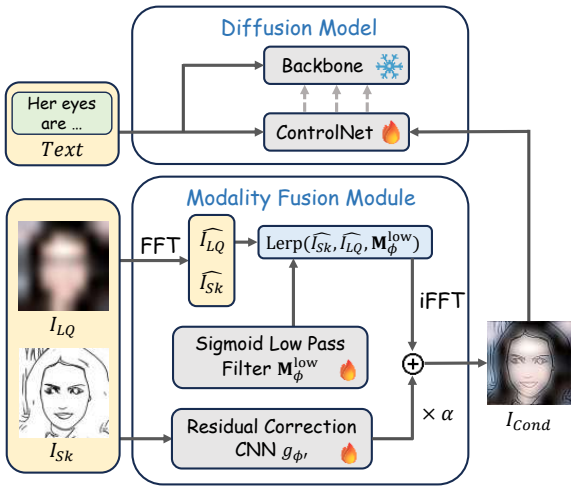
To support the training of IdentiFace, existing public datasets require significant adaptation. Standard face datasets often lack the paired multi-modal annotations (fine-grained text, simple sketch, low-quality images) necessary for our framework. In this section, we detailed the construction of our specialized datasets ID-CelebA and ID-FFHQ. Table 1 lists their basic information. Images are saved as .jpg considering disk space, but .png images are still available in original datasets and can be processed step-by-step accordingly. All data processing complies with the original licenses and ethical guidelines of the source datasets.

Table 1. Basic information of ID-CelebA and ID-FFHQ.

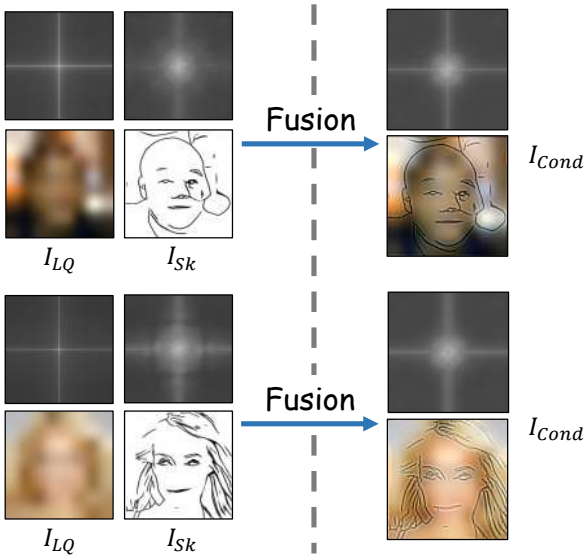
Names	ID-CelebA	ID-FFHQ
Derived From	CelebAMaskHQ [6]	FFHQ [4]
Image Count	18,197	25,839
Identity Count	5,054	21,857
Resolutions	512×512	1024×1024
Attributes	Text [1], Sketch [11, 12], I_{LQ}, I'_{LQ}	
Saved Form	.jpg	

A. Quality and Pose Filtering

To prevent ambiguity in conditional generation, each image must contain a single, clearly visible face. Multi-face images in FFHQ (few in CelebAMask-HQ) are excluded to ensure one-to-one mapping between conditions and outputs. Furthermore, frontal faces are preferred as they contain the most discriminative biometric features. We employed a SoTA Vision-Language Model (VLM) *Qwen3-VL* [1] to automatically filter images based on face count



(a) Visualization of our modality fusion module. LQ image I_{LQ} and sketch image I_{Sk} contain low and high-frequency information respectively. The uni-modal conditional input I_{Cond} is a mixture of the two images in frequency domain, added by a residual correction CNN. I_{Cond} and text are then passed to the diffusion model with ControlNet architecture as conditions.



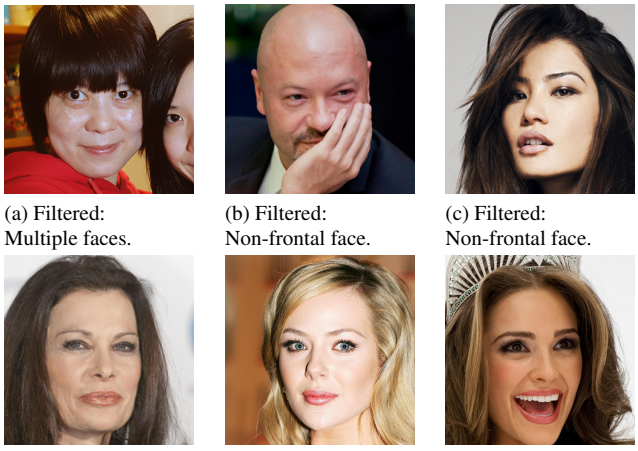
(b) Visualization of frequency maps between I_{LQ}, I_{Sk} and I_{Cond}. The fusion is conducted by ID-PA per latent-space channel.

Figure 1. Demonstration of fusion module and frequency analysis.

and pose orientation with appropriate prompt engineering. Fig. 2 shows examples of filtered and kept images.

B. Accessory Removal

In forensic identification, accessories (e.g., eyeglasses, hats, masks) often occlude key facial landmarks and are typically absent in standard identity photos (e.g., mugshots or ID cards). Training on images with accessories may cause the model to rely on non-biometric cues, degrading



(a) Filtered: Multiple faces. (b) Filtered: Non-frontal face. (c) Filtered: Non-frontal face.

(d) Kept. (e) Kept. (f) Kept.

Figure 2. Examples of filtered and kept images in our datasets.

identification performance. We utilized *SegFace* [8], a facial semantic segmentation model, to detect salient accessories. These regions were inpainted using image editing model *Qwen-Image-Edit* [14] to restore the underlying facial structure. Subsequently, *Qwen3-VL* was used again to quality-check the edited images, removing those with significant artifacts or quality degradation.

Table 2. Image Removal Summary. "AdaFace Disabled" means AdaFace model fails to detect face from the image; "VLM Blockage" refers to that VLM APIs block output due to content review.

Removal Reasons	ID-CelebA	ID-FFHQ
Multiple Faces	N/A	13154
Non-frontal Face	11,620	29,131
Edit Degradation	60	236
Incorrect Cluster	0	1,615
AdaFace Disabled	18	17 + 1,615
VLM Blockage	105	8
Remaining	18,197	25,839

C. Identity Clustering and Verification

Since the official FFHQ [4] dataset lacks an identity file, we constructed identity clusters in ID-FFHQ using facial recognition model *AdaFace* [5], with a pre-calculated threshold $T_{Ada} \approx 0.385$.

Applying this threshold to ID-FFHQ revealed a large cluster of children's faces (1,615 images) incorrectly grouped together, indicating reduced recognition reliability on low-age demographics. To maintain high identity purity, this cluster was excluded. Additionally, all images where faces could not be detected by *AdaFace* were removed.

D. Attribute Synthesis Details

For textual descriptions, we used *Qwen3.5-VL* to fine-grained descriptions by a prompt-engineered template. For



(a) Original Image with glasses.



(b) Inpainted Image.



(c) Original Image with hat.



(d) Inpainted Image.



(e) Original Image with glasses and hat.



(f) Inpainted Image, removed due to degraded quality.

Figure 3. Examples of inpainted images in our datasets.

simple sketch images, we leveraged a pretrained GAN model [11, 12] with the processing scripts of MM-CelebA-HQ [15] to generate sketches from GT images. For I'_{LQ}, i.e., LQ closer to raw surveillance images, we added more camera-like degradations, including optical blurring and realistic noise injection, with a higher resized resolution of 32×32 (24×24 for real-world experiment samples), compared to 8×8 for I_{LQ}. Fig. 4 and Fig. 5 demonstrate examples of GT images with conditional attributes in our dataset.



(a) GT image.



(b) Sketch Image.



(c) camera-like LQ.

Figure 4. Example of an image with conditional image attributes.



(a) GT image.



(b) Sketch Image.



(c) camera-like LQ.

Figure 5. Example of another image with conditional image attributes.

III. Additional Info on Experiments

A. Additional Setup

Table 3 shows the detailed experiment setups for model instances we trained and compared, where *PixArt-ControlNet-Based* represents ControlNet-PA (LQ), ControlNet-PA (Sketch), and ID-PA; and *SD1.5-ControlNet-Based* refers to ControlNet-SD (LQ), ControlNet-SD (Sketch), and ID-SD; both with various input conditions. Note that ID-PA (zero text) uses the same weights as ID-PA, and ControlNet-PA (Dual Branch) uses the weights for sketch and LQ. Note that ablations on $\mathcal{L}_{ID}$ and LTR are not taken into account in Table 3.

Both *Dual Branch* models (ControlNet-PA and T2I-Adapter) use separately trained weights on LQ and sketch, and add the calculated features together to inject into the model backbone, with ratios of *0.5, 0.5* and *1, 1*, which produce feasible results with both conditional controls. They might not be optimal values, but results are sufficient to demonstrate that our modality fusion method surpasses simply adding branches.

For those models with $\mathcal{L}_{ID}$, the additional loss function is linearly added to the existing ControlNet loss function defined by their official implementations, with a scaling factor defined as a hyperparameter. $\mathcal{L}_{ID}$ is calculated every 10 training steps. The total training loss can be defined as

$$\mathcal{L}_{total} = \mathcal{L}_{Ctrl} + \alpha \cdot \mathcal{L}_{ID}(\text{step}\%10 == 0), \tag{5}$$

where $\mathcal{L}_{Ctrl}$ is the existing ControlNet loss, and $\alpha = 0.1$ is the scaling hyperparameter.

B. More Synthetic Experiment Results

In our paper, we introduced local text ratio (LTR) for textual input diversity. A similar design is *unconditional guidance* (UCG), meaning that text condition would be dropped out with a fixed probability p_u during training. LTR takes place only when the text is not dropped out. We did not provide explicit UCG when training the aforementioned models, and here we conducted ablations on UCG with $p_u = 0.1$ and $p_u = 0$ (existing) for ID-SD. Quantitative results in Table 4 shows that the model performs slightly better when UCG is employed.

Table 3. Detailed experiment setup for each model and baseline we trained or compared in modality selection analysis. a/b denotes values used in ID-CelebA / ID-FFHQ, and a single value means used in both datasets. "40e" refers to 40 epochs and "100,000i" refers to 100,000 training iterations.

Models	Text	LQ	Sketch	Backbones	Batch Sizes per GPU	GPU Counts	Learning Rates	Epochs or Iterations	LTRs	$\mathcal{L}_{ID}$
PixArt-ControlNet-Based [2, 3]	various			PixArt-α [3]	16/20	various	2e-5	40e/30e	0.3	✓
SD1.5-ControlNet-Based [10, 16]	various			SD v1.5 [10]	24	1	1e-5	40e/30e	0.3	✓
OSDFace[13]	✗	✓	✗	SD v2.1 [10]	pretrained					
DiffFaceSketch[9]	✗	✗	✓	VQ-F4 [10]	12	1	3e-5	100,000i	N/A	✗
T2I-Adapter[7]	✓	✓	✓	SD v1.5 [10]	16	1	1e-4	40e/30e	0	✗

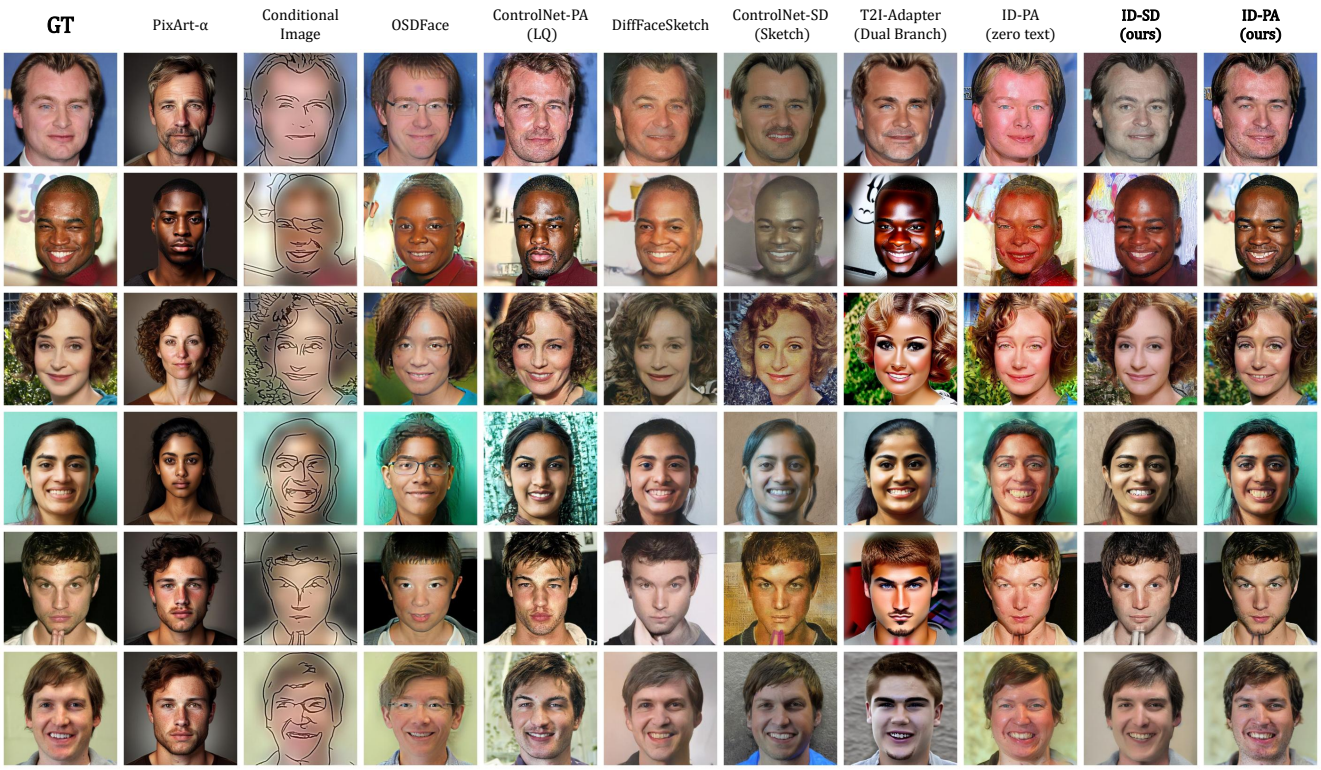


Figure 6. Qualitative comparison of different modality selections and baselines under one-shot generation.

Table 4. Ablation studies on UCG values for ID-SD, trained and tested on dataset ID-CelebA.

Iter. Gen.	p_u	AF. Match Rate↑	Avg. AF. Sim↑	SF. Match Rate↑	Avg. SF. Sim↑
✗	0	0.66482	0.30911	**0.65604**	**0.56981**
	0.1	**0.67054**	**0.31068**	0.65440	0.56904
✓	0	0.83718	0.33189	**0.72772**	**0**.57695
	0.1	**0.84471**	**0.33293**	0.72412	**0.57736**

Furthermore, Fig. 6 demonstrates qualitative comparisons between IdentiFace and other methods. Fig. 7 and Fig. 8 contain examples of simulated iterative processes of ID-PA and ID-SD. SegFace [8] provides semantic segmentation masks on areas corresponding to textual descriptions. The mask for editing is the union of masked areas of GT and the generated image to be edited.

C. Analysis of Real-world Experiments

As the synthetic iterative generation examples show, some fine-grained facial feature manipulations may not be sufficiently noticeable to the human eye. To facilitate facial feature reconstructions in real-world scenarios, we enable users to additionally edit sketch and random seed in every iteration, and *Undo* after the first generation is allowed, compared with oppositions in the simulated user policy.

We configured and conducted our real-world experiment in the way mentioned in our main paper and above, featuring 20 additional training epochs on ID-FFHQ containing I'_{LQ} for model ID-PA (pretrained on ID-FFHQ for 30 epochs). Experiment results show high pass rates, while

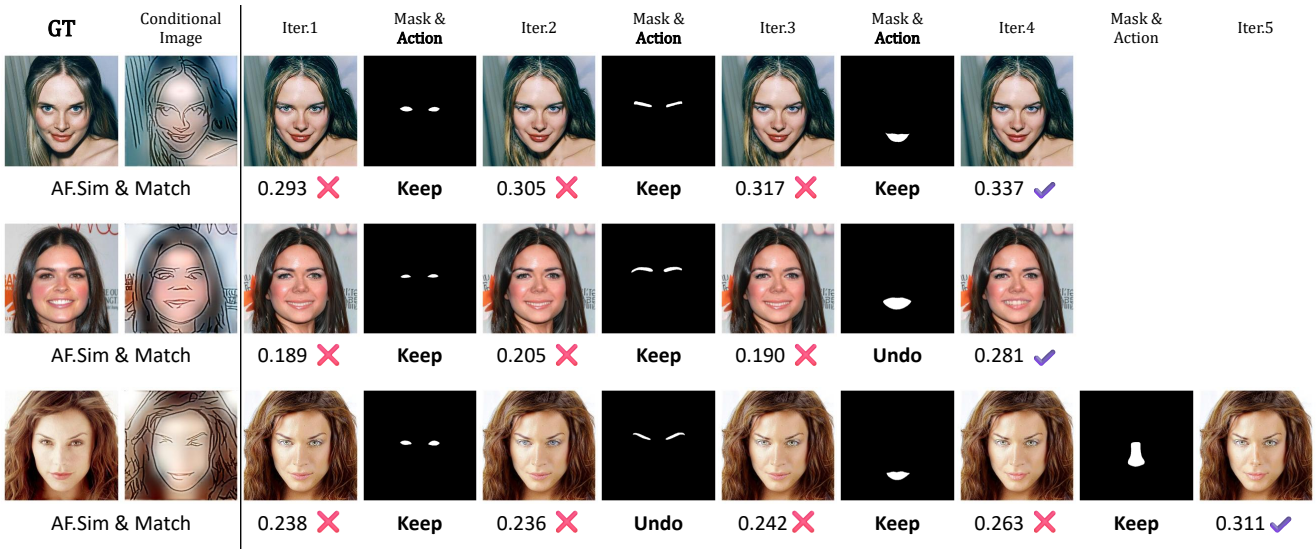


Figure 7. Iterative generation process demonstration for ID-PA. After each iteration, the pipeline automatically choose from *Return, Keep, Undo* according to AdaFace similarity and identity matching conditions. if *Undo* is chosen, editing of the next iteration will be based on image in the last iteration with action *Keep*.

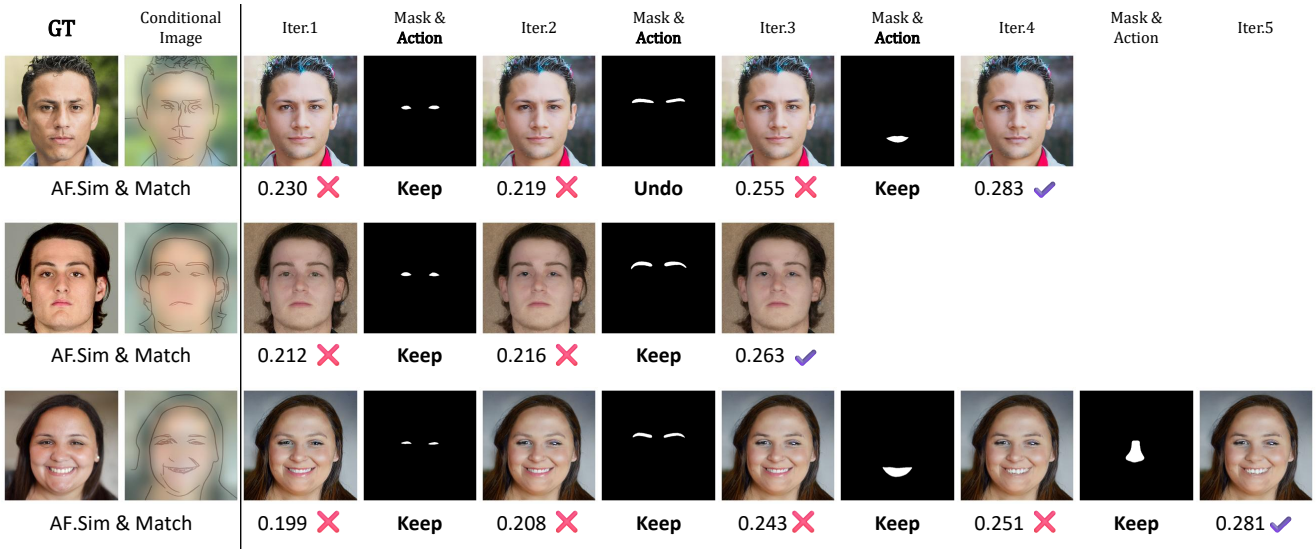


Figure 8. Iterative generation process demonstration for ID-SD.

average time and iteration counts vary in different samples (Table 5). They are mainly influenced by testing orders, sample difficulties, and volunteers' strategy preferences. For instance, S2 was an easy sample and was often tested early in our experiments, when volunteers were less familiar with the system. As a result, although they succeeded in only one or two iterations, they spent considerable time in the first iteration writing textual descriptions and drawing sketches, which contributed to a longer total trial time per iteration. S5 and S6 are challenging samples. Among the successful trials, volunteers passed S5 in fewer iterations and less time by providing appropriate initial text and sketches. In contrast, those who passed S6 repeatedly refined the input conditions based on the model's output, revealing a strategic difference in their approaches.

Beyond mask-guided editing, users have additional manipulation choices. A correct identity can be retrieved by repeatedly adjusting random seeds and sketches for conditional generation, merely *Undo* the misidentified generated image. Fig. 9 and 10 show two real examples where using the *Undo* operation led to successful results.

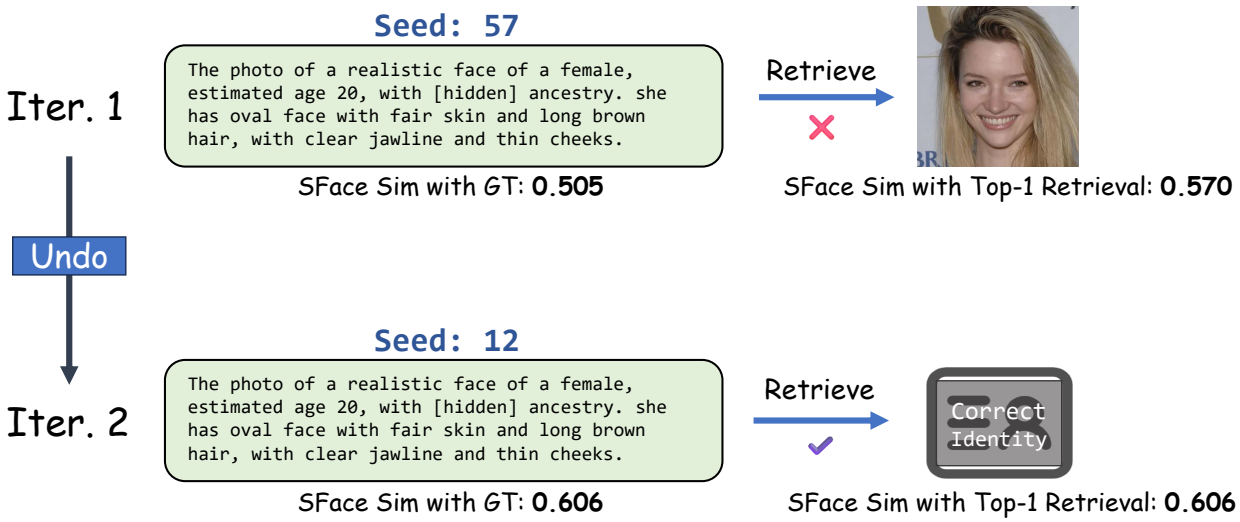


Figure 9. Example of an iterative generation process using solely *Undo* in our real-world experiment. Some information is hidden due to privacy concern.

Table 5. Per-sample results of real-world experiments.

Metrics	S1	S2	S3	S4	S5	S6
Total Trials	4	4	5	3	4	4
Passed Trials	4	4	5	3	2	2
Avg. Iter.	4.00	1.25	3.20	6.33	3.00	5.50
Avg. Time	9.4	6.2	5.6	12.1	3.0	10.2

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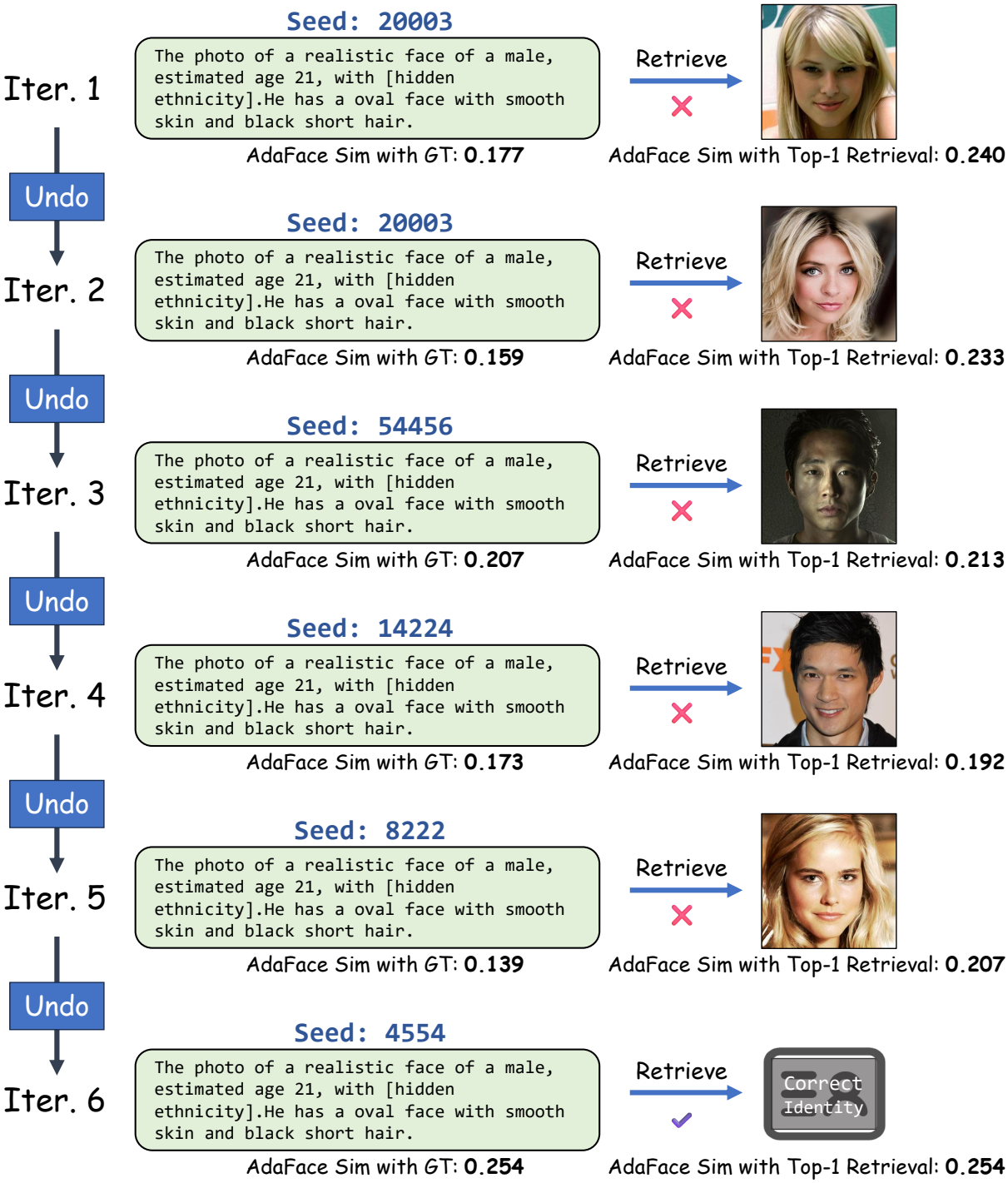


Figure 10. Another example of an iterative generation process using solely *Undo* in our real-world experiment.

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